YoungPyo Hong

Graduate Student in Astronomy, Seoul National University
evan77333@gmail.com · hongyp007@snu.ac.kr
ORCID: 0009-0002-6238-2326 · GitHub: hongyp007 · Website: hongyp007.github.io

EDUCATION

Integrated M.S./Ph.D. Program in Astronomy Sep. 2025 – present
Seoul National University, Department of Physics and Astronomy
Advisor: Prof. Myungshin Im · **GPA:** 3.55 / 4.3 · **M.S. expected:** Aug. 2027
Thesis (in progress): optical afterglows of gamma-ray bursts

B.S. in Physics Mar. 2020 – Feb. 2023
Yonsei University, Mirae Campus
GPA: 4.05 / 4.3 · graduated one year early
- Ranked **first in the department** at graduation
- Admitted as the **top-ranked student** of the entering class

RESEARCH INTERESTS

- **Time-domain astronomy**: gamma-ray bursts, kilonovae, and gravitational-wave counterparts
- **Multi-wavelength** and **multi-messenger** follow-up of transient events
- **Machine learning** and **AI** applications in astronomy

PUBLICATIONS

Paek, G. S. H., Im, M., Jeong, M., Chang, S.-W., Hur, M. M., **Hong, Y. P.**, Kim, S., Lee, J., Lee, D., & Lee, S.-H. 2025, "GECKO Follow-up Observations of the Binary Neutron Star–Black Hole Merger Candidate S230518h," *The Astrophysical Journal*, 981, 38

CONFERENCE PRESENTATIONS

International

Cho, J., et al., incl. **Hong, Y. P.** 2022, "Performance and Application of the Radio Telescope in Gwacheon National Science Museum," *IAU General Assembly*, Busan, Korea [talk]

Domestic

Hong, Y. P., Tak, D., Im, M., Kim, J. H., Chang, S.-W., Paek, G. S. H., Choi, H., Hyun, D., & Lee, W.-H. 2026, "Multi-band Optical SED Analysis of GRB 251013C Afterglow with the 7-Dimensional Telescope," *Korean Astronomical Society Spring Meeting*, Suwon, Korea [talk]

Hong, Y. P., Im, M., Choi, H., Kim, S., & Paek, G. S. H. 2026, "Automated Wide-Field Rapid Follow-up System for Transient Events with TCSpy and gpPy," *Instrumentation for Astronomy and Space Science (IASS)*, Yeosu, Korea [talk]

Hong, Y. P., Im, M., Moon, H., Kim, J. H., & Choi, C. 2025, "Filter Transmittance Characterization for 15 New Medium-Band Filters of the 7-Dimensional Telescope: Measurement Procedures and Validation Results," *IASS*, Osong, Korea [poster]

RESEARCH EXPERIENCE

Graduate Researcher — Transient Follow-up with 7DT and RASA36 Sep. 2025 – present
Seoul National University · Advisor: Prof. Myungshin Im
- Built an **end-to-end automated rapid target-of-opportunity (ToO) system** for RASA36, running from alert reception through transient search by integrating gpPy (photometry pipeline) and TCSpy (telescope operation)
- Operate and maintain the system for **remote wide-field follow-up** of transient alerts
- Multi-band optical **SED analysis of gamma-ray burst afterglows** (e.g. GRB 251013C) with 7DT

Post-Baccalaureate Research Intern Sep. 2024 – Aug. 2025
Seoul National University, Department of Physics and Astronomy · Advisor: Prof. Myungshin Im
- Joined the 7DT program on transient phenomena and multi-messenger astronomy
- Characterized the **transmittance of 15 new medium-band 7DT filters** and validated the measurement procedure

Undergraduate Research — Remote Radio Observation Nov. 2021 – Feb. 2022
Gwacheon National Science Museum radio telescope
- Proposal: "Comparison between the distance of radio-loud quasars and their radio power"
- Analyzed 18 hours of observations; no significant radio signal was detected

OBSERVING EXPERIENCE

RASA36 — automated ToO system, implementation and operation Sep. 2025 – present
El Sauce Observatory, Chile · remote observing
- Operate the automated ToO system I built; remote follow-up of **gamma-ray burst and gravitational-wave alerts**

7-Dimensional Telescope (7DT) Sep. 2025 – present
El Sauce Observatory, Chile · remote observing
- Target-of-opportunity follow-up of **gamma-ray bursts and gravitational-wave alerts**

AWARDS AND HONORS

- Next Generation of Researchers in Basic Sciences Scholarship, Seoul National University — Feb. 2026
- Honorable Mention, Science and Technology Convergence Academic Seminar, Yonsei University — Dec. 2022
- Honorable Mention, Department of Physics Academic Seminar, Yonsei University — Nov. 2022
- Yonsei Vision Scholarship, Yonsei University — Sep. 2022
- Truth Scholarship (half tuition, academic excellence), Yonsei University — Sep. 2022
- Academic Honors, Yonsei University — Feb. 2022
- National Program of Excellence in Software Scholarship, Yonsei University — Mar. 2021
- Yonsei Special Scholarship (full tuition; admission with highest distinction) — Mar. 2020

SCHOOLS AND WORKSHOPS

- **Astro AI Asian Network (A3N) Summer School** (2024, 2025) — statistical modeling and deep learning for astronomy
- **Summer / Winter School on Numerical Relativity and Gravitational Waves** (2022, 2023, 2024) — GW detection principles and data analysis
- **KIAS Summer School on Extragalactic Astronomy and Cosmology** (2023, 2024) — hydrodynamic and N-body simulations, SED fitting, CMB data processing
- **ALMA Summer School** (2026) — ALMA data reduction and polarization analysis
- **ALMA Proposal Preparation Workshop** (2025) — proposal: “Environment of merger events with short GRB afterglows at sub-mm”
- **K-GMT Summer School** (2023) — Gemini/GNIRS proposal: “Further insight on galaxy rotation curves at high redshift with GNIRS”
- **Radio Summer School** (2022) — receiver systems, interferometry, AGN, SKA
- **Workshops:** SKA-Korea (2024, 2025), Korea Rubin Science (2025), LSST with AI (2025), Woojeon (2025, 2026), 10th Anniversary of the Discovery of Gravitational Waves (2025), Extragalactic Transients (2026)

TEACHING AND OUTREACH

Teaching Assistant, Seoul National University Sep. 2025 – Dec. 2026
Man and the Universe (Fall 2025, Fall 2026); Astronomical Observation and Laboratory II (Spring 2026)

Astronomy Communicator, Level II, Korean Amateur Astronomical Society 2022
Trained in public outreach and in presenting astronomical concepts to non-expert audiences.

Physics Tutor for First-Year Undergraduates, Yonsei University Mar. – Jun. 2022

Korean University Knowledge Mentoring Program Apr. 2021 – Jan. 2022
Academic tutoring and career counseling for elementary, middle, and high school students.

MILITARY SERVICE

Republic of Korea Army — mandatory military service Apr. 2023 – Jul. 2024